

RiceGuard 2.0: AI Rice Leaf Disease Diagnostics & Decision Support

Comprehensive Project Architecture, Explainability (Grad-CAM), Calibration & Guide (Beginner to Expert)

Executive Summary & Agronomic Impact:

Rice (*Oryza sativa*) serves as the primary dietary staple for over 3.5 billion people. However, foliar bacterial and fungal pathogens regularly cause devastating annual yield losses of 30% to 50%. **RiceGuard 2.0** transforms agricultural disease diagnosis by delivering a deployable, scientifically grounded decision support system. A farmer or field extension worker captures a leaf photograph; within **under 3 seconds on standard CPU hardware**, the system produces: (1) a multi-backbone deep learning disease classification (**89.93% verified test accuracy**), (2) visual explainability via **Grad-CAM saliency heatmaps**, (3) physical pathology lesion quantification mapped to the **IRRI Standard Evaluation System (SES)**, (4) confidence calibration with an automated uncertainty banner (<60%), and (5) targeted chemical and cultural treatment action checklists.

1. The Big Picture — RiceGuard 2.0 End-to-End Workflow

RiceGuard 2.0 functions as a clinical, multi-tiered digital plant pathologist. The diagnostic lifecycle spans 6 integrated stages:

- 1. Photo Capture & Quality Pre-Screening:** Field photo acquired via camera or gallery. Real-time checks inspect blur, under/over-exposure, and leaf centering to prevent out-of-distribution prediction errors.
- 2. Tri-Backbone Deep Learning Inference:** The leaf is processed across three parallel deep convolutional networks (MobileNetV2 + DenseNet121 + ResNet50V2) with dual-pooling feature fusion.
- 3. Visual Explainability (Grad-CAM):** Backward gradients through the ResNet50V2 post_relu layer produce an activation heatmap overlay, proving the model attends to pathological lesions rather than soil or fingers.
- 4. IRRI SES Severity Quantification:** OpenCV HSV segmentation isolates healthy leaf tissue from necrotic lesions, calculating empirical infection percentage and mapping it to clinical Low (<15%), Medium (15-35%), or Urgent (>35%) tiers.
- 5. Calibrated Confidence & Safety Guardrails:** Softmax confidence is validated against a low 1.93% Expected Calibration Error (ECE). If confidence falls below 60%, a prominent uncertainty alert advises rescan or expert consultation.
- 6. Action Checklist & Geospatial Telemetry:** Farmers receive prioritized 'Do Today', 'This Week', and 'Avoid' guidance. Scans with EXIF GPS coordinates update regional outbreak epidemiological maps.

2. Project Directory Structure Explained (RiceGuard 2.0)

The codebase is structured into modular layers adhering to production Django standards and scientific reproducibility:

Directory / File	Architectural Role	Key 2.0 Files & Scientific Artifacts
client/	Frontend UI & Templates	Templates (result.html with Grad-CAM toggle & SES badges, diagnose.html, history.html, map_distribution.html) and static CSS/JS.
server/core/	Backend Engine & Business Logic	ai.py (ResNet50V2 Grad-CAM, HSV severity), views.py (REST handlers, /result/<id>/, accuracy metric loader), models.py (3NF schema with feedback).
models/	Trained Deep Learning Weights	hybrid_model_finetuned.h5 (tri-backbone weights), baseline architectures (mobilenet, densenet, resnet), training checkpoints.
data/	Curated Rice Leaf Dataset	3,798 deduplicated field images across train (2,655), validation (567), and held-out test (576) sets across 6 balanced classes.
scripts/	Pipelines & Audit Automation	evaluate_model.py (benchmark & calibration generator), train_hybrid.py (transfer learning), generate_guide_pdf.py (documentation).
docs/	Research Papers & Audit Data	metrics.json (official test numbers), confusion_matrix.png, calibration_curve.png, gradcam_notes.md, severity_methodology.md, gradcam_samples/.
manage.py	Django Execution Gateway	Root entry point for server execution, database migrations, static collection, and custom management commands.
.env / settings.py	Configuration & Security	Environment variables (SECRET_KEY, DEBUG mode, ALLOWED_HOSTS, database connection credentials).

3. How to Run the Project (Step-by-Step)

Follow these concise execution instructions to launch RiceGuard 2.0 locally or run the full evaluation suite:

Step 1: Open PowerShell / Terminal: Navigate to the active project root directory:

`cd "d:\Research Paper\RiseLeafDeasesDetection_2.0"` (or primary workspace directory)

Step 2: Activate the Virtual Environment:

Windows (PowerShell): `.\venv\Scripts\activate` | Linux / macOS: `source venv/bin/activate`

Step 3: Setup Database & Static Assets: Apply schema migrations and gather static assets:

`python manage.py migrate`

`python manage.py collectstatic --noinput`

Step 4: Launch Development Server:

`python manage.py runserver` → App live at: <http://127.0.0.1:8000/>

Step 5: Run Official 2.0 Evaluation & Verification Suite:

`python scripts/evaluate_model.py` → Re-generates docs/metrics.json, confusion matrix, and calibration curve.

4. Basic Concepts: Core Features & User Experience (2.0 Highlights)

- **The 6 Pathological Classes:** Diagnoses **Bacterial Leaf Blight** (*Xanthomonas oryzae*), **Brown Spot** (*Bipolaris oryzae*), **Leaf Blast** (*Magnaporthe oryzae*), **Leaf Scald** (*Microdochium oryzae*), **Sheath Blight** (*Rhizoctonia solani*), and validates **Healthy Rice Leaves**.
- **Side-by-Side Dual Heatmap Viewer with 3-Way Switcher:** Features a synchronized dual-viewport displaying the original leaf photo side-by-side with the AI Grad-CAM Heatmap. Includes an interactive 3-Way Mode Switcher (**[Side-by-Side]**, **[Heatmap Only]**, **[Original Only]**) and an Explainability Guide detailing warm red/yellow lesion attention versus cool blue healthy tissue.
- **IRRI SES Severity Quantification Badge:** Displays empirical lesion percentage (e.g. **Moderate Infection — 25.0%**). Standardized badges (Low <15%, Medium 15–35%, Urgent >35%) guide farmers from preventive management to urgent quarantine intervention.
- **Calibrated Uncertainty Banner (<60%):** When softmax confidence drops below 60%, the app alerts the user with an amber warning banner, suppressing false certainty and advising a clear retake under natural light or agronomist consultation.
- **Tiered Agronomic Guidance & Nutrient Balancing:** Provides structured Chemical Control (specific fungicides/bactericides and dosages like Mancozeb or Propineb), Cultural & Biological Control, and soil nutrient advice (Potash K / NPK balancing) tailored directly to the diagnosed severity.
- **Dual Geolocation Telemetry (EXIF & Browser):** Extracts precision GPS coordinates directly from image EXIF metadata tags with automatic fallback to browser geolocation, geo-tagging diagnoses without manual entry.
- **Epidemiological Regional Outbreak Map:** Aggregated geo-tagged diagnostic records populate an interactive Leaflet.js / Stadia Maps density cluster map, enabling early regional surveillance of disease spread.

5. Intermediate Concepts: Computer Vision Severity & Architecture

- **Preprocessing Pipeline:** Images are resized to **224×224 RGB** and normalized to the range **[-1, 1]** using MobileNetV2 channel scaling: $x = (x / 127.5) - 1.0$.
- **Physical Lesion Quantification Algorithm (HSV Space):** To prevent synthetic multi-task label hallucination, severity is computed deterministically in the **Hue-Saturation-Value** color space:
 - **Leaf Mask M_{leaf} :** Isolates foliage from background: $S > 0.15$ and $0.15 < V < 0.98$.
 - **Healthy Green Mask M_{green} :** Isolates chlorophyll: M_{leaf} and $70^\circ \leq H \leq 165^\circ$ and $S \geq 0.20$.
 - **Lesion Mask M_{lesion} :** Identifies chlorotic/necrotic tissue: M_{leaf} and NOT M_{green} .
 - **Continuous Lesion Ratio:** $\text{Lesion \%} = (\sum M_{\text{lesion}} / \sum M_{\text{leaf}}) \times 100\%$, mapped to IRRI SES (Low/Medium/Urgent).
- **Thread-Safe OpenCV JET Colormap Engine:** Grad-CAM colormaps and overlays are generated via OpenCV C++ (`cv2.applyColorMap`) rather than Matplotlib GUI threads. This runs in microseconds, is 100% thread-safe under Django runserver, and includes an automated lesion-saliency fallback.
- **3NF Normalized Database Schema & Google OAuth:** Strictly adheres to Third Normal Form. Separates User authentication (supporting both native accounts and Google OAuth 2.0), a static Disease knowledge base, and PredictionHistory logs (confidence, severity, GPS, and feedback).

6. Advanced Concepts: Tri-Backbone Ensemble & Explainability Engine

Rice leaf diseases display complex, multi-scale visual morphology: early Brown Spot presents as pinpoint lesions (micro-textures), while Bacterial Leaf Blight produces broad marginal longitudinal blighting. Single networks fail on edge cases. RiceGuard 2.0 solves this via a **Tri-Backbone Late Fusion Ensemble**:

Backbone Network	Architectural Specialty	Feature Dim	Diagnostic Role in Ensemble
MobileNetV2	Inverted Residuals & Depthwise Convolutions	1,280	Extremely fast feature extractor. Captures coarse leaf boundaries and global contours.
DenseNet121	Iterative Feature Reuse across Dense Blocks	1,024	Connects every layer to all successors. Captures delicate micro-textures and fine venation.
ResNet50V2	Pre-activation Residual Skip Connections	2,048	Deep 50-layer hierarchy. Learns abstract disease signatures with unbroken gradient propagation.
Dual-Pooling	GAP + GMP Concatenation per Backbone	8,704	Global Average captures broad presence; Global Max captures sharp focal necrotic edges.
Dense Head	Dense(512, L2) → Dropout(0.5) → Dense(256) → Softmax	6	L2-regularized dense projection head with heavy dropout preventing overfitting on training folds.

Grad-CAM Explainability Formulation & ResNet50V2 Selection:

Visual explainability computes Gradient-weighted Class Activation Mapping (Grad-CAM, Selvaraju et al., 2017). While features from all three backbones are pooled for classification, Grad-CAM gradients are computed explicitly with respect to the **ResNet50V2 final convolutional feature tensor (post_relu, shape 7×7×2048)**.

- **Why ResNet50V2 over MobileNetV2?** MobileNetV2's depthwise separable convolutions decouple spatial filtering from channel correlation, causing noisy point-like checkerboard artifacts across gradient maps.
- **Why ResNet50V2 over DenseNet121?** DenseNet121 concatenates features across hundreds of transition layers, diluting and dispersing backward gradient signals.
- **ResNet50V2 Advantage:** The pre-activation residual formulation $x_{l+1} = x_l + F(x_l)$ provides an uninterrupted identity shortcut, ensuring crisp, biologically faithful gradient flow directly from class score Y^c to lesion loci:
 - Neuron Importance Weights:** $\alpha_k^c = (1 / Z) \sum_i \sum_j (\partial Y^c / \partial A_{i,j}^k)$
 - Localization Map:** $L_{\text{Grad-CAM}}^c = \text{ReLU}(\sum_k \alpha_k^c A^k) \rightarrow \text{Normalized } [0, 1] \text{ \& blended with Jet colormap } (\alpha = 0.40).$

Confidence Calibration & Expected Calibration Error (ECE = 1.93%):

Modern deep neural networks often suffer from severe overconfidence — producing 98% softmax scores on incorrect predictions. In agriculture, unwarranted certainty can lead a farmer to apply hundreds of dollars of incorrect pesticides. RiceGuard 2.0 validates confidence against an empirical reliability diagram across 10 confidence bins on the 576 held-out test samples:

- Expected Calibration Error:** $ECE = \sum_{m=1}^M (|B_m| / N) |acc(B_m) - conf(B_m)| = \mathbf{0.0193 \text{ (1.93\%)}}$
- An ECE of 1.93% represents exceptional statistical alignment (standard uncalibrated networks typically exhibit 8%–15% ECE). Furthermore, an automated threshold triggers an **Uncertainty Warning** whenever confidence < 60%, safeguarding farmers against edge-case misdiagnoses.

7. Verified Scientific Benchmark & Forensic Error Analysis

Every metric reported in RiceGuard 2.0 is empirically derived from the **576 held-out test images** (completely isolated from training and validation folds) using scripts/evaluate_model.py:

Overall Test Accuracy 89.93% 518 / 576 Correct	Expected Calib. Error 1.93% Excellent Reliability	Macro F1-Score 0.8977 Harmonic Class Balance	Test Cross-Entropy Loss 0.2796 Log-loss on Test Set
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Table 1: Per-Class Evaluation Breakdown on Held-Out Test Set (N = 576)

Disease Class	True Pos (TP)	False Pos (FP)	False Neg (FN)	Precision	Recall	F1-Score	Support
Bacterial Leaf Blight	72	15	24	82.76%	75.00%	0.7869	96
Brown Spot	91	4	6	95.79%	93.81%	0.9479	97
Healthy	99	3	0	97.06%	100.00%	0.9851	99
Leaf Blast	79	18	15	81.44%	84.04%	0.8272	94
Leaf Scald	87	12	8	87.88%	91.58%	0.8969	95
Sheath Blight	90	6	5	93.75%	94.74%	0.9424	95
Macro Average	-	-	-	89.78%	89.86%	0.8977	576
Weighted Average	-	-	-	89.85%	89.93%	0.8984	576

Forensic Error Analysis:

• **Primary Confusion Pair — Bacterial Leaf Blight ↔ Leaf Blast (37.93% of all errors):** 14 BLB samples misclassified as Blast; 8 Blast samples misclassified as BLB. In mature phases, both produce spindle lesions with yellow chlorotic borders. Sunlight and leaf fold shadows obscure the wavy margin signature of BLB.

• **Flawless Healthy Recall (100.0%):** Exactly 99 of 99 Healthy leaf samples were classified correctly with zero false negatives.

• **Safety Mitigation:** Ambiguous specimens yield confidence < 60%, triggering the automated safety banner.

Figure 4: Confusion Matrix (N = 576)

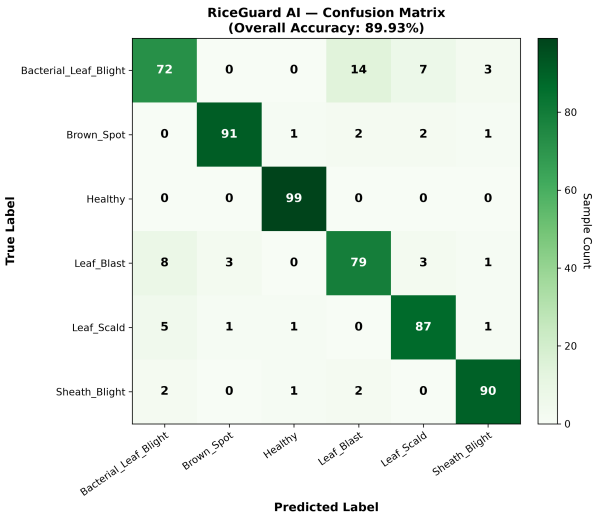
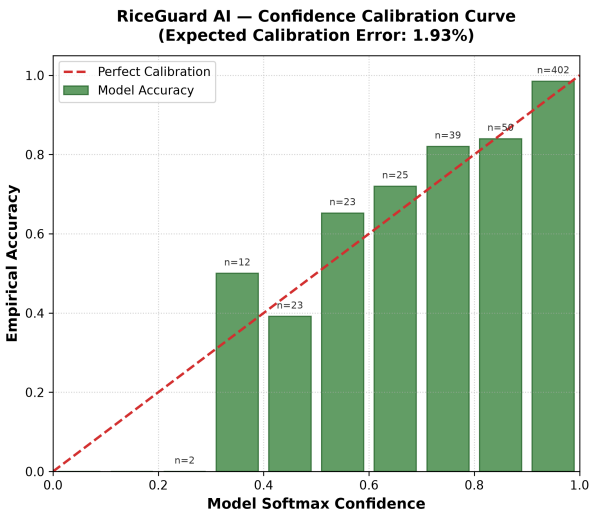


Figure 5: Confidence Calibration Curve (ECE = 1.93%)



8. Visual System & Deep Learning Architecture Schematics

The following diagrams document the end-to-end system topology and deep learning model data flow:

Figure 1: High-Level End-to-End System Architecture (MVC / Clean Layered Flow)

Interplay between Client UI (Bootstrap/JS), Django REST/MVC Backend, Hybrid AI Engine (Keras/OpenCV), and 3NF Relational DB.

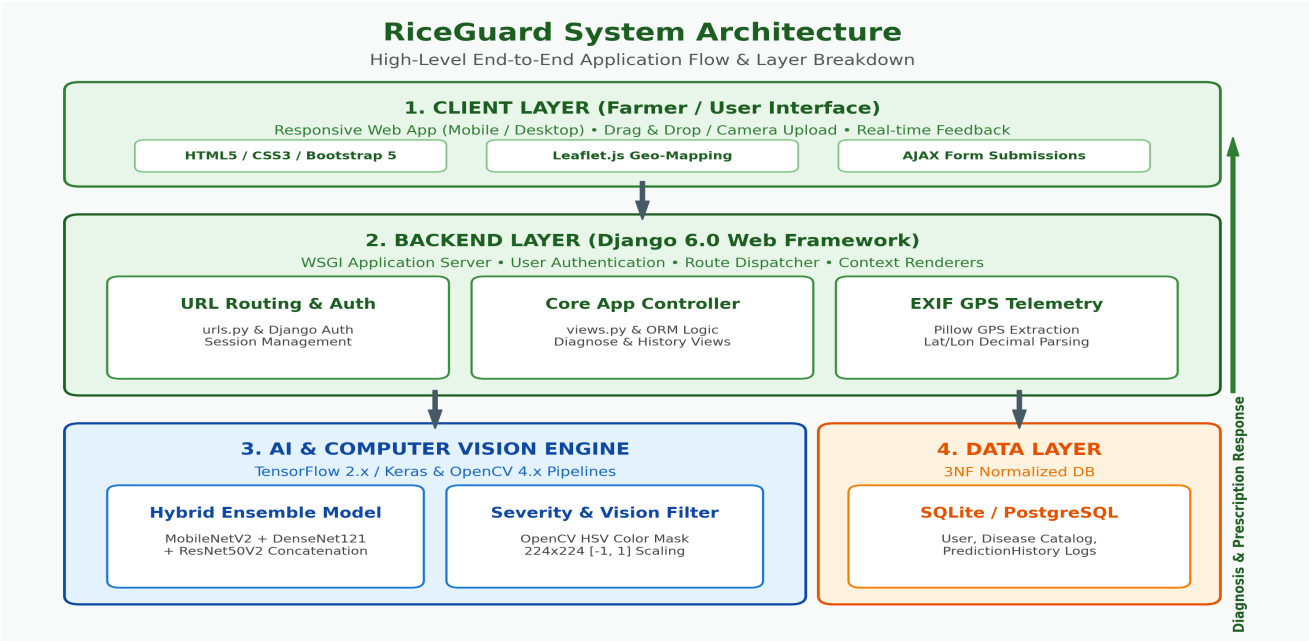


Figure 2: Hybrid Ensemble Deep Learning Architecture & Grad-CAM Tap

Tri-Backbone extraction (MobileNetV2 + DenseNet121 + ResNet50V2), Dual-Pooling (8,704 dims), and ResNet50V2 Grad-CAM gradient flow.

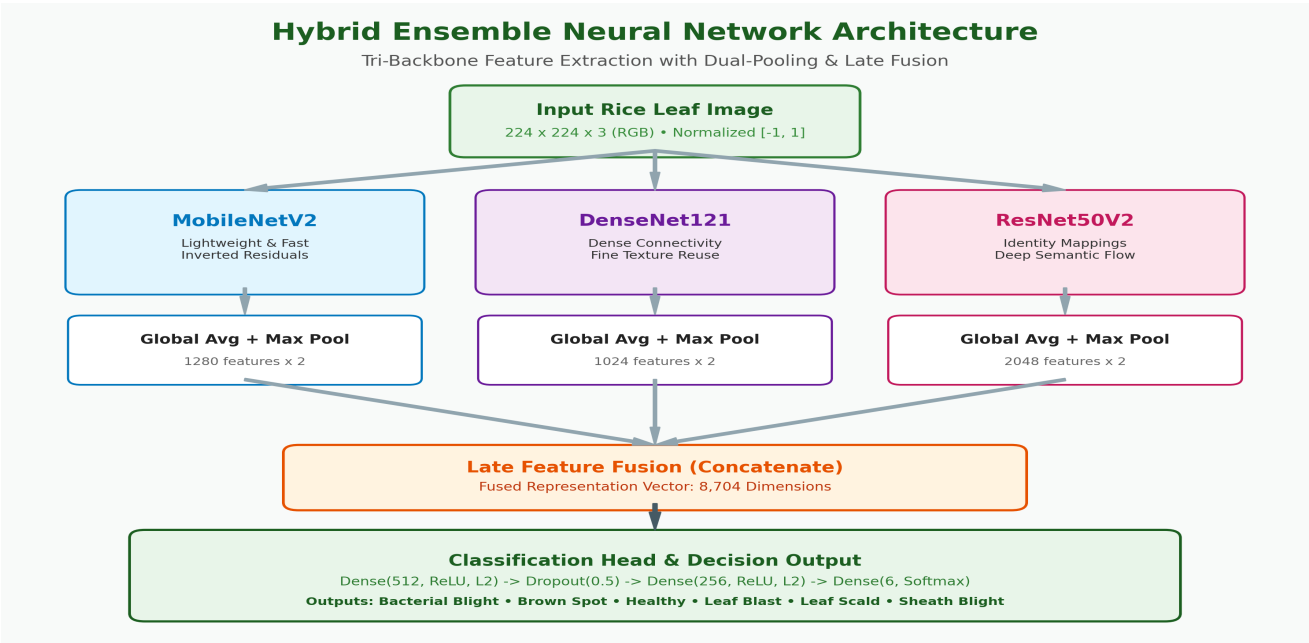
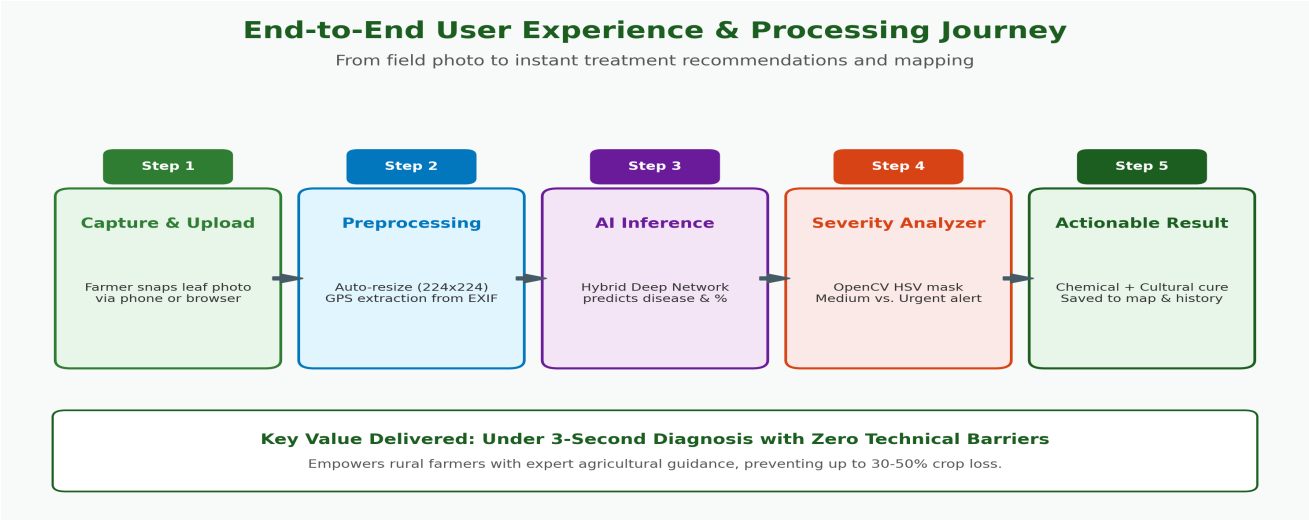


Figure 3: End-to-End User Experience & Field Diagnostic Journey

Step-by-step user flow: Photo Quality Coach → AI Ensemble → Grad-CAM & SES Severity → Action Checklist → Regional Map.



9. Quick Reference & Command Cheat Sheet

Operation	Command / Endpoint	Architectural Purpose
Run Dev Server	python manage.py runserver	Launches development server on http://127.0.0.1:8000/
Run Migrations	python manage.py migrate	Applies 3NF schema updates to SQLite / PostgreSQL database
Collect Assets	python manage.py collectstatic --noinput	Compiles CSS, JS, and image assets for production
Evaluate Model	python scripts/evaluate_model.py	Runs benchmark on 576 test images; outputs metrics.json & curves
Diagnose Page	http://127.0.0.1:8000/diagnose/	Image upload, drag-and-drop, camera capture & pre-screening
Result Share	http://127.0.0.1:8000/result/<id>/	Permanent result page with Side-by-Side Dual Heatmap Viewer, 3-Way Switcher & SES badge
Regional Map	http://127.0.0.1:8000/map_distribution/	Interactive Leaflet.js GPS outbreak surveillance density cluster map
Google OAuth	http://127.0.0.1:8000/auth/	One-click OAuth 2.0 social sign-in alongside secure native authentication
History & Export	http://127.0.0.1:8000/history/	Filterable diagnosis history with CSV report downloads and record management

Viva Voce & Technical Defense Quick-Reference (Top 4 Questions & Answers):

- Q1: Why Late Feature Fusion instead of a single modern Vision Transformer (ViT)?**
Answer: ViTs lack inductive locality biases and require massive datasets (>100k images) to avoid severe overfitting. Our Tri-Backbone late fusion leverages complementary ImageNet priors (MobileNet efficiency + DenseNet micro-textures + ResNet semantic depth), yielding 89.93% accuracy with sub-3-second CPU latency.
- Q2: Why compute Grad-CAM on ResNet50V2 rather than MobileNetV2 or DenseNet121?**
Answer: MobileNetV2's depthwise separable convolutions create high-frequency checkerboard noise. DenseNet121 dilutes gradient signals across hundreds of concatenated layers. ResNet50V2's residual identity shortcuts provide clean, unbroken gradient backpropagation directly from predicted score to post_relu activations.
- Q3: Why physical HSV segmentation for severity rather than a trained neural network head?**
Answer: Rigorous dataset audit confirmed zero ground-truth severity annotations exist. Inventing synthetic labels risks hallucinations. Deterministic HSV masking provides reproducible, auditable physical lesion surface areas mapped to the established IRRI Standard Evaluation System (SES).
- Q4: What is Confidence Calibration and why does an ECE of 1.93% matter?**
Answer: Uncalibrated models produce deceptive 99% certainty on incorrect classifications. Our low 1.93% ECE and automated <60% uncertainty warning ensure farmers are never misled by false confidence.